



Marcelo Bongarti

Ph.D. in Applied Mathematics

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Weierstraß Institute for Applied Analysis and Stochastics (WIAS), Berlin, Germany

Research Interests

Partial differential equations (PDE-constrained optimization, boundary control and stabilization of PDEs, parabolic and hyperbolic equations); (generalized) Nash equilibrium problems (GNEPs) and game theory; stochastic calculus and quantitative/financial modeling; applications to digital twin technology; scientific computing in Python.

Employment

- 06/2021–present** **Postdoctoral Researcher**
Weierstraß Institute for Applied Analysis and Stochastics — Berlin, Germany
Projects: *Berlin Cluster of Excellence Math+ (AA4-3)*; *SFB-TRR 154 (B02, C08)*.
- 05/2020–08/2020** **NSF–MSGI Summer Research Intern**
University of California, Los Angeles — Los Angeles, CA, USA
Mentor: Andrea L. Bertozzi.
- 05/2019–08/2019** **NSF–MSGI Summer Research Intern**
Argonne National Laboratory — Chicago, IL, USA
Mentors: Paul Hovland and Sri H. K. Narayanan.

Education

- 2016–2021** **Ph.D., Mathematics**
University of Memphis — Memphis, TN, USA (GPA 3.99)
Thesis: *On nonlinear acoustic waves of third-order (in time) and applications*. Advisors: Irena Lasiecka & Roberto Triggiani.
- 2014–2016** **M.Sc., Mathematics**
São Paulo State University — S.J. Rio Preto, Brazil (GPA 3.88)
Thesis: *Domains of Fractional Powers of Operators*. Advisor: German Lozada–Cruz.
- 2010–2014** **B.Sc., Mathematics**
São Paulo State University — S.J. Rio Preto, Brazil

Awards & Achievements

- 2021** Morton Thesis Award (nominated), University of Memphis
- 2020** Dr. Ralph Faudree Research Award, University of Memphis; NSF–MSGI Summer Research Internship, UCLA
- 2019** Student Research Forum Award for Best Poster, University of Memphis; NSF–MSGI Summer Research Internship, Argonne National Laboratory
- 2014** National Journey of Scientific Initiation, IMPA
- 2006–09** Brazilian Olympiad of Mathematics for Public Schools (Honor Mention)

Publications

Preprints & Submitted Work

- 2026** A. Alphonse, M. Bongarti, E. Valdinoci. *A free boundary problem driven by boundary distance in the coincidence set*. [arXiv:2607.21355](https://arxiv.org/abs/2607.21355).
- 2026** M. Bongarti, M. Hintermüller. *State constrained convex Nash equilibrium problems coupled with linear hyperbolic PDEs*. arXiv preprint [arXiv:2607.13299](https://arxiv.org/abs/2607.13299). doi:[10.48550/arXiv.2607.13299](https://doi.org/10.48550/arXiv.2607.13299)
- 2026** M. Bongarti, M. Hintermüller. *Structure versus regularity of set-valued maps in convex generalized Nash equilibrium problems in Banach spaces*. arXiv preprint [arXiv:2512.12831](https://arxiv.org/abs/2512.12831). doi:[10.48550/arXiv.2512.12831](https://doi.org/10.48550/arXiv.2512.12831)
- 2025** Z. Amer, A. Avdzhieva, M. Bongarti, P. Dvurechensky, P. Farrell, U. Gotzes, F. M. Hante, A. Karsai, S. Kater, M. Liero, et al. *Modeling hydrogen embrittlement for pricing degradation in gas pipelines*. WIAS Preprint No. 3201. doi:[10.20347/WIAS.PREPRINT.3201](https://doi.org/10.20347/WIAS.PREPRINT.3201)

Journal Articles

- 2025** M. Bongarti, C. Parkinson, W. Wang. *Optimal control of a reaction-diffusion epidemic model with non-compliance*. European Journal of Applied Mathematics, 1–26. doi:10.1017/S0956792525000130
- 2025** R. C. Contreras, M. S. Viana, E. S. Fonseca, M. A. dos S. Bongarti, Ö. Toygar, R. C. Guido. *Exploring multicepstral features in a new classical machine learning-based framework for replay attack detection*. Computers & Electrical Engineering, 127, 110570. doi:10.1016/j.compeleceng.2025.110570
- 2023** M. Bongarti, M. Hintermüller. *Optimal boundary control of the isothermal semilinear Euler equation for gas dynamics on a network*. Applied Mathematics and Optimization, 89(36). doi:10.1007/s00245-023-10088-0
- 2022** A. Bertozzi, M. Bongarti, L. D. Galvan, L. Hatcher, M. R. Lindstrom, C. Parkinson, C. Wang. *Alternative SIAR models for infectious diseases and applications in the study of non-compliance*. Mathematical Models and Methods in Applied Sciences, 32(10), 1987–2015. doi:10.1142/S0218202522500464
- 2022** M. Bongarti, I. Lasiecka. *Boundary feedback stabilization of a critical nonlinear JMGT equation with Neumann-undissipated part of the boundary*. Discrete and Continuous Dynamical Systems - S, 15(8), 1957–1985. doi:10.3934/dcdss.2022107
- 2022** M. Bongarti, I. Lasiecka, J. H. Rodrigues. *Boundary stabilization of the linear MGT equation with partially absorbing boundary data and degenerate viscoelasticity*. Discrete and Continuous Dynamical Systems - S, 15(6), 1355–1376. doi:10.3934/dcdss.2022020
- 2021** M. Bongarti, I. Lasiecka, R. Triggiani. *The SMGT equation from the boundary: regularity and stabilization*. Applicable Analysis, 101(5), 1735–1773. doi:10.1080/00036811.2021.1999420
- 2020** M. Bongarti, S. Charoenphon, I. Lasiecka. *Vanishing relaxation time dynamics of the Jordan Moore-Gibson-Thompson equation arising in nonlinear acoustics*. Journal of Evolution Equations. doi:10.1007/s00028-020-00654-2

Conference Proceedings

- 2024** R. C. Contreras, G. L. Heck, M. S. Viana, M. A. dos S. Bongarti, H. Zamani, R. C. Guido. *Metaheuristic algorithms for enhancing multicepstral representation in voice spoofing detection: an experimental approach*. International Conference on Swarm Intelligence, 247–262. doi:10.1007/978-981-97-7181-3_20
- 2024** R. C. Contreras, M. S. Viana, P. C. Pessoa, M. Bongarti, H. Zamani, R. C. Guido. *Enhancing metaheuristic algorithms for voice spoofing detection via multi-cepstral representation*. LNCS 14822, Springer, 247–262. doi:10.1007/978-3-031-84356-3_19
- 2024** M. S. Viana, R. C. Contreras, P. C. Pessoa, M. A. dos S. Bongarti, H. Zamani, R. C. Guido, O. Morandin Junior. *Massive conscious neighborhood-based crow search algorithm for the pseudo-coloring problem*. International Conference on Swarm Intelligence, 182–196. doi:10.1007/978-981-97-7181-3_15
- 2022** S. H. K. Narayanan, T. Propson, M. Bongarti, J. Hückelheim, P. Hovland. *Reducing memory requirements of quantum optimal control*. Computational Science – ICCS 2022, Springer, 129–142. doi:10.1007/978-3-031-08760-8_11
- 2021** M. Bongarti, I. Lasiecka, J. H. Rodrigues. *Optimal boundary feedback control of high intensity focused ultrasound processes*. 60th IEEE Conference on Decision and Control (CDC), 3318–3323. doi:10.1109/CDC45484.2021.9683592

Books & Chapters

- 2021** M. Bongarti, I. Lasiecka. *Boundary stabilization of the linear MGT equation with feedback Neumann control*. Deterministic and Stochastic Optimal Control and Inverse Problems, 7, 150–168. doi:10.1201/9781003050575
- 2020** M. Bongarti, S. Charoenphon, I. Lasiecka. *Singular thermal relaxation limit for the Moore-Gibson-Thompson equation arising in propagation of acoustic waves*. Semigroups of Operators – Theory and Applications, vol. 325, Springer, 147–182. doi:10.1007/978-3-030-46079-2_9

Other

- 2021** M. Bongarti. *On Nonlinear Acoustic Waves of Third-Order (in Time) and Applications*. Ph.D. thesis, University of Memphis. Electronic Theses and Dissertations Archive, 2518. digitalcommons.memphis.edu/etd/2518
- 2021** M. Bongarti, G. Lozada-Cruz. *Alguns Teoremas do Tipo Valor Médio: de Lagrange a Malesevici*. Revista Matemática Universitária, 1, 17. doi:10.21711/26755254/rmu20215
- 2016** M. Bongarti. *Domains of Fractional Powers of Operators*. Master's thesis, São Paulo State University (UNESP). Advisor: German Lozada-Cruz.

Organizational Activities

- 2024** Organizer, mini-symposium *Modeling, Analysis and Optimal Control of Infinite Dimensional Problems and Applications*, IFIP TC7 Conference, Hamburg.
- 2021** Local organizer, 15th International Conference on Free Boundary Problems: Theory and Applications (FBP 2021), virtual.

Teaching & Mentoring

- 2023** **Weierstraß Institute for Applied Analysis and Stochastics**
Berlin, Germany
 ▶ Mentor (alongside Prof. Michael Hintermüller) of Ms. Evaelle Riffard at the “WIAS internship program” (Summer)
- 2021** **Department of Mathematical Sciences**
University of Memphis, Memphis, TN, USA
 ▶ Mentor at the “NSF–STEM collaborative research initiative” (Spring)
- 2020** **Department of Mathematical Sciences**
University of Memphis, Memphis, TN, USA
 ▶ Independent instructor for “Differential and Integral Calculus I” (Spring)
 ▶ Responsible for the media preparation for “Differential and Integral Calculus III” (Spring)
 ▶ Independent instructor for “Differential and Integral Calculus I” (Fall)
 ▶ Instructor (joint work with Prof. Irena Lasiecka) for “Distribution Theory and Sobolev Spaces” (Fall)
 ▶ Mentor at the “NSF–STEM collaborative research initiative” (Spring and Fall)
- 2019** **Department of Mathematical Sciences**
University of Memphis, Memphis, TN, USA
 ▶ Independent instructor for “Ordinary Differential Equations” (Spring)
 ▶ Mentor at the “NSF–STEM collaborative research initiative” (Summer)
 ▶ Independent instructor for “Differential and Integral Calculus I” (Fall)
- 2018** **Department of Mathematical Sciences**
University of Memphis, Memphis, TN, USA
 ▶ Independent instructor for “Elementary Calculus” (Fall)
 ▶ Independent instructor for “Elementary Calculus” (Summer)
 ▶ Responsible for the “Analysis PhD Qualifying Exam” Review Sessions (Summer)
- 2017** **Department of Mathematical Sciences**
University of Memphis, Memphis, TN, USA
 ▶ Independent instructor for “Trigonometry” (Fall)
- 2015** **Istitute of Biosciences, Languages and Exact Sciences**
São Paulo State University, S.J. Rio Preto, SP, Brazil
 ▶ Teaching assistant for “Differential and Integral Calculus I” (Fall)

Selected Talks

- 05/2026** *Poster:* MOG 2026 – Mathematics of Networks with Physical Transport, Nürnberg, Germany.
- 03/2025** *Invited:* “Multi-objective optimization with linear hyperbolic PDE constraints: generalized Nash equilibrium problems and gas market applications,” WIAS Nonsmooth Variational Problems Seminar, Berlin.
- 07/2023** *Invited:* SIAM Conference on Control and Its Applications (CT23), Philadelphia, PA.
- 06/2023** *Invited:* 93rd GAMM Annual Meeting, TU Dresden, Germany.
- 11/2022** *Invited:* Oberseminar PDGs, Universität Konstanz, Germany.
- 10/2022** *Invited, online:* Early Career Math Colloquium, University of Arizona.
- 07/2022** *Invited:* Waves 22, ENSTA Paris, and IFIP-TC7, Warsaw (two talks).
- 10/2020** *Invited mini-course:* Web Mathematics Week, São Paulo State University; Dr. Ralph Faudree Lectures, University of Memphis.
- 2017–20** Additional invited/contributed talks at the University of Memphis, São Paulo State University, and Carnegie Mellon University (full list available on request).

Skills

- Languages** English (fluent), Brazilian Portuguese (native).
- Coding** $\text{\LaTeX}$; Python (basic).

Miscellaneous

- Reviewer** SIAM J. Mathematical Analysis; ESAIM: COCV; Acta Mechanica; J. Optimization Theory & Applications; Discrete & Continuous Dynamical Systems; Applied Mathematics & Optimization.
- Certificates** IBM Data Science with Python (Coursera, ongoing); Machine Learning (Coursera); SQL (CodeInfinity).

References

Prof. Irena Lasiecka, University of Memphis — lasiecka@memphis.edu

Prof. Roberto Triggiani, University of Memphis — rtrggani@memphis.edu

Prof. Andrea Bertozzi, University of California, Los Angeles — bertozzi@math.ucla.edu

Prof. German Lozada-Cruz, São Paulo State University — german.lozada@unesp.br